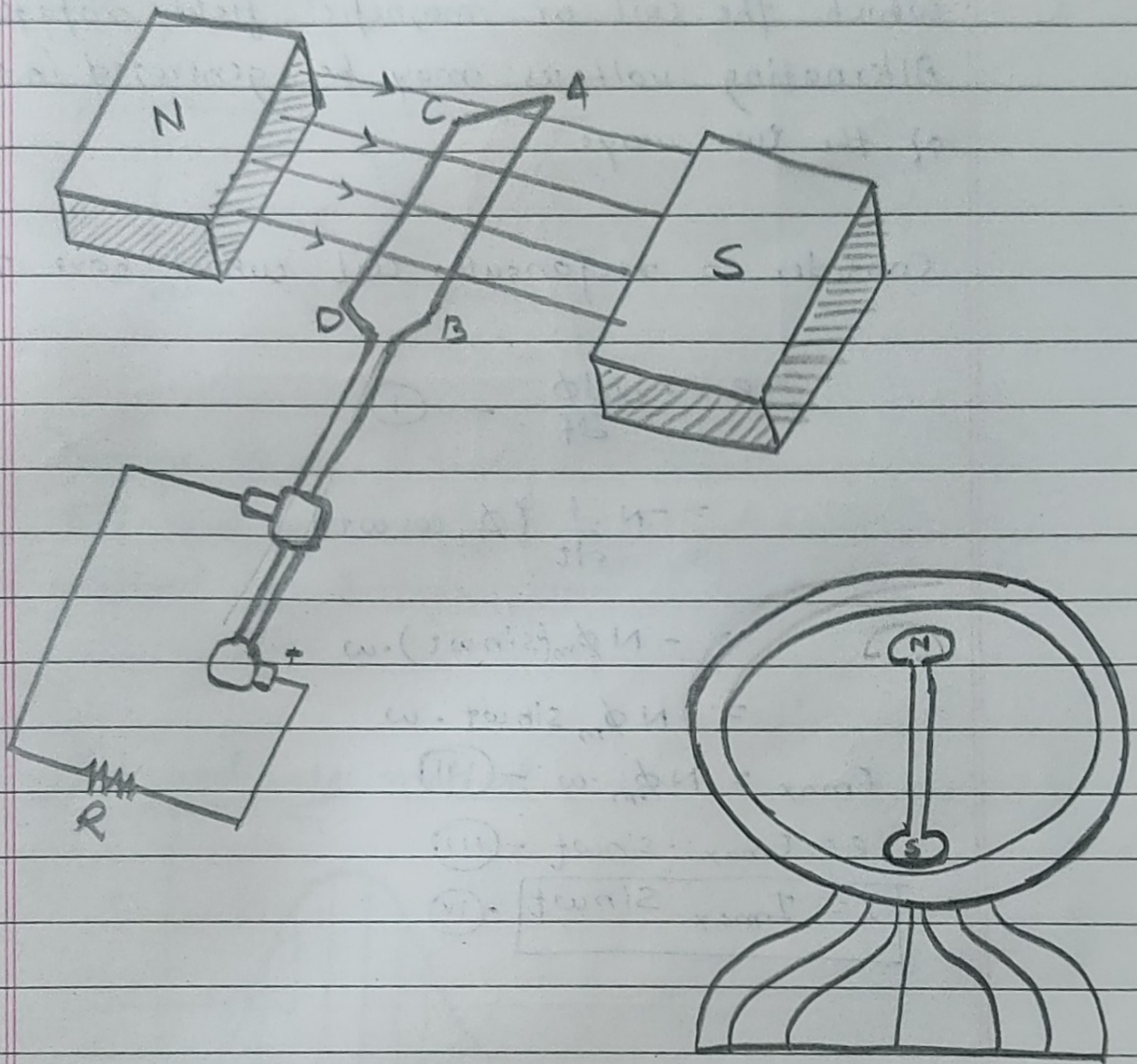
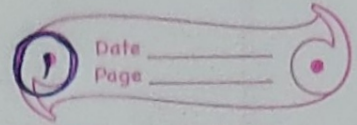


## AC Fundamentals



Alternating voltages may be generated by rotating a coil in a magnetic field or by rotating a magnetic field within a stationary coil. The value of the voltage generated depends on in each case upon the no. of turns of the coil, strength of the field and the speed at

which the coil or magnetic field rotates. Alternating voltages may be generated in either of the two ways.

Consider a rectangular coil which have  $n$  turns

$$e = -N \frac{d\phi}{dt} \quad \text{--- (i)}$$

$$= -N \frac{d}{dt} (\phi_m \cos \omega t)$$

$$= -N \phi_m (\sin \omega t) \cdot \omega$$

$$= N \phi_m \sin \omega t \cdot \omega$$

$$E_{\max} = N \phi_m \cdot \omega \quad \text{--- (ii)}$$

$$e = E_{\max} \cdot \sin \omega t \quad \text{--- (iii)}$$

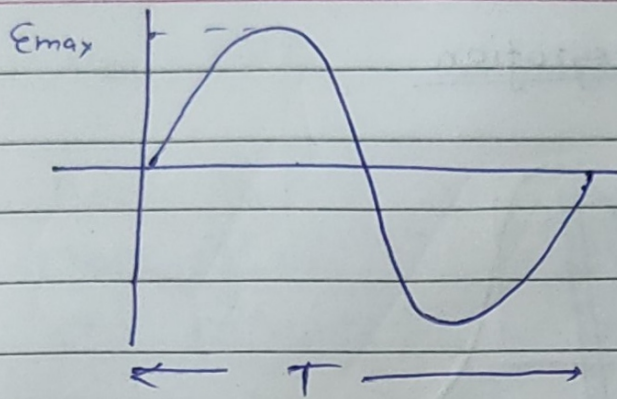
$$\boxed{I = I_{\max} \sin \omega t} \quad \text{--- (iv)}$$

Cycle:

One complete set of positive and negative values of alternating quantity is known as cycle.

Time period

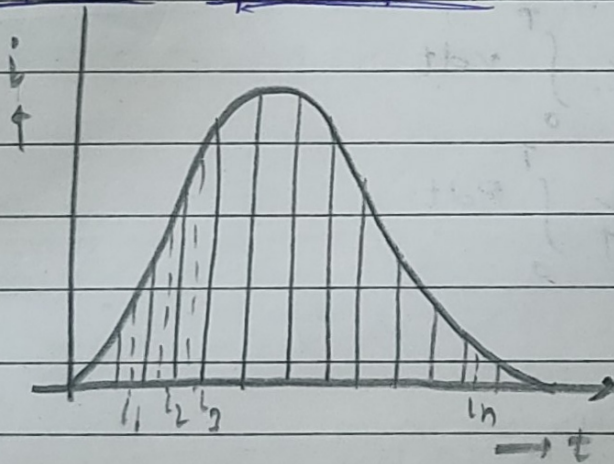
The time taken by an alternating quantity to complete one cycle is called time period



## // . Average value

- (i) Mid ordinate method
- (ii) Method of Integration.

### (i) Mid ordinate method.

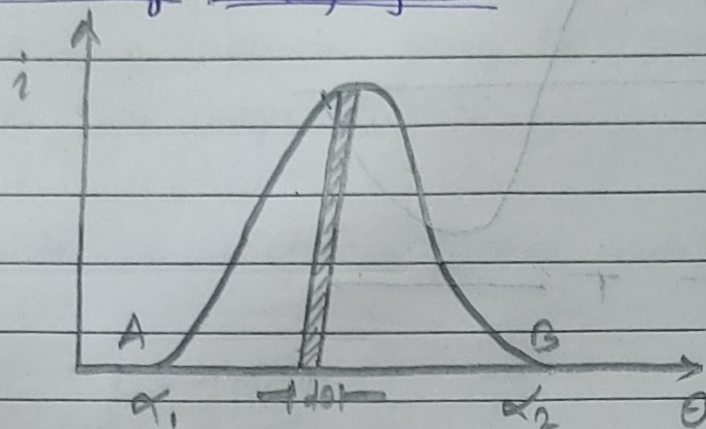


$A \cdot V = \frac{\text{area under the Curve } (i_1 + i_2 + \dots + i_n)}{\text{base or length of curve.}}$

Average value.

$$= \frac{i_1 + i_2 + i_3 + \dots + i_n}{n}$$

## (ii) Method of Integration



$$A.V = \frac{1}{\alpha_2 - \alpha_1} \int_{\alpha_1}^{\alpha_2} i d\theta$$

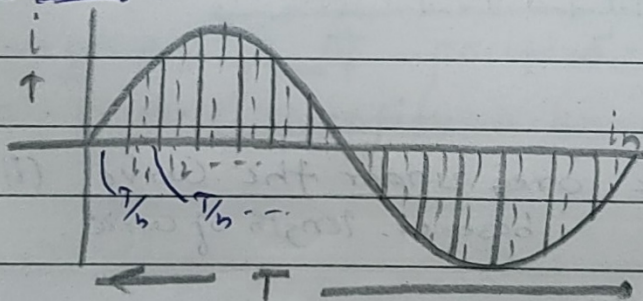
$$I_{av} = \frac{1}{T} \int_0^T i dt$$

$$V_{av} = \frac{1}{T} \int_0^T v dt$$

$$P_{av} = \frac{1}{T} \int_0^T P dt$$

### • RMS value

Mid ordinate method



$$H_1 = i_1^2 RT/n$$

$$H_2 = i_2^2 RT/n$$

$$H_n = i_n^2 RT/n$$

$$H_{ac} = H_1 + H_2 + \dots + H_n$$

$$= i_1^2 RT/n + i_2^2 RT/n + \dots + i_n^2 RT/n$$

$$H_{DC} = I^2 RT$$

$$H_{DC} = H_{AC}$$

$$I^2 RT = (i_1^2 + i_2^2 + i_3^2 + \dots + i_n^2) \frac{RT}{n}$$

$$I^2 = \frac{(i_1^2 + i_2^2 + i_3^2 + \dots + i_n^2)}{n}$$

$$I_{rms} = \sqrt{\frac{i_1^2 + i_2^2 + i_3^2 + \dots + i_n^2}{n}}$$

Square root of the mean of square of current

- The effective or RMS value of an alternating current is that steady current (DC) which ~~when~~ flowing through a given resistor for a given time produces the same amount of heat as produce by alternating current when flowing through the same resistance (resistor) for the same time

RMS value by integration

$$H_{AC} = \int_0^T i^2 R dt$$

$$H_{DC} = I^2 RT$$

$$H_{AC} = H_{DC}$$

$$I^2 RT = \int_0^T i^2 R dt$$

$$I^2 = \frac{1}{T} \int_0^T i^2 dt$$

$$I = \sqrt{\frac{1}{T} \int_0^T i^2 dt}$$

$$V = \sqrt{\frac{1}{T} \int_0^T v^2 dt}$$

Form factor ( $k_f$ )

It is defined as the ratio of rms value of the wave and average value of the wave.

$$\text{form factor} = \frac{\text{rms value}}{\text{avg value}}$$

Amplitude factor ( $k_a$ )

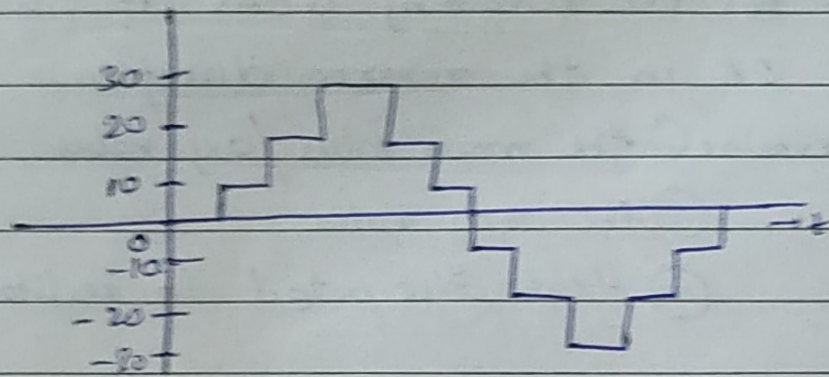
$$k_a = \frac{\text{Peak value}}{\text{rms value}} \quad \text{or} \quad \frac{\text{max value}}{\text{rms value}}$$

$$I = I_0 \sin \theta$$

$$I_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T i^2 dt}$$

$$= \sqrt{\frac{1}{2A} \int_0^{2A} I_m^2 \sin^2 \theta d\theta}$$

Q A current has the following steady values in ampere for equal intervals of time changing suddenly from one value to the next  
0, 10, 20, 30, 20, 10, 0, -10, -20, -30, -20, -10, 0



Calculate avg value, rms value, form factor and peak factor

$$\begin{aligned} I_{av} &= \frac{1}{6} (i_1 + i_2 + i_3 + i_4 + i_5 + i_6) \\ &= \frac{1}{6} (90) \\ &= \underline{\underline{15 \text{ A}}} \end{aligned}$$

$$\begin{aligned} I_{rms} &= \sqrt{\frac{1}{6} (0^2 + 10^2 + 20^2 + 30^2 + 20^2 + 10^2 + 0^2)} \\ &= 17.8 \text{ A} \end{aligned}$$

$$\text{Form factor } (K_f) = \frac{17.8}{15}$$

$$\text{Amplitude factor } (K_a) = \frac{30}{17.8}$$

Q A current varying with a periodic wave form flows through an  $8\ \Omega$  resistor in accordance with the following pattern

- (i) for the first  $0.04\text{ s}$  it is constant at  $6\text{ A}$
- (ii) For the next  $0.02\text{ s}$  it is zero
- (iii) for the remaining  $0.04\text{ s}$  it is constant at  $2\text{ A}$  in the reverse direction.

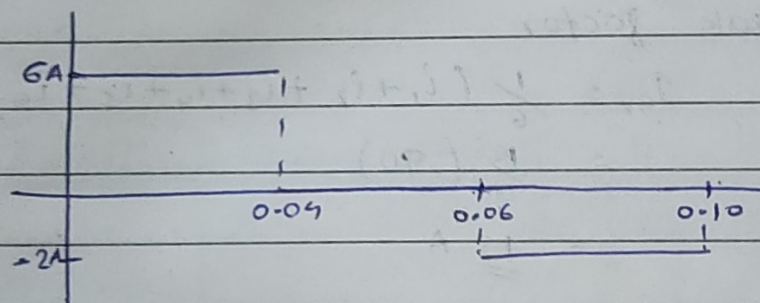
Determine (i) the mean value / avg value

(ii) the rms value

(iii) Heat dissipated in resistor in  $5\text{ min}$ .

$$\textcircled{1} \text{ mean value} = \frac{6 \times 0.04 + 0 \times 0.02 + (-2) \times 0.04}{0.10} = \frac{0.24 + 0 - 0.08}{0.10}$$

$$= \frac{0.16}{0.10} = \frac{16}{10} = 1.6\text{ A}$$



$$\textcircled{2} \text{ rms value} = \sqrt{\frac{6^2 \times 0.04 + 0^2 \times 0.02 + (-2)^2 \times 0.04}{0.10}}$$

$$= \sqrt{\frac{36 \times 0.04 + 0 + 4 \times 0.04}{0.10}}$$

$$= \sqrt{\frac{1.44 + 0.16}{0.10}}$$

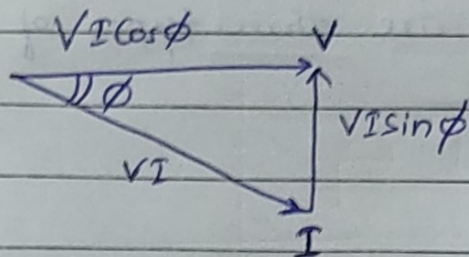
$$= \sqrt{\frac{1.60}{0.10}} = \sqrt{\frac{160}{10}} = 4\text{ A}$$

③  $H = I_{rms}^2 \times R \times T$  \*

$$H = 4^2 \times 8 \times 5 \times 60$$

$$= 38400 \text{ J}$$

$$= 38.4 \text{ kJ}$$



$P = VI \cos \phi \rightarrow \text{watt, kW}$

$Q = VI \sin \phi \rightarrow \text{kVA}_r, \text{VA}_r$

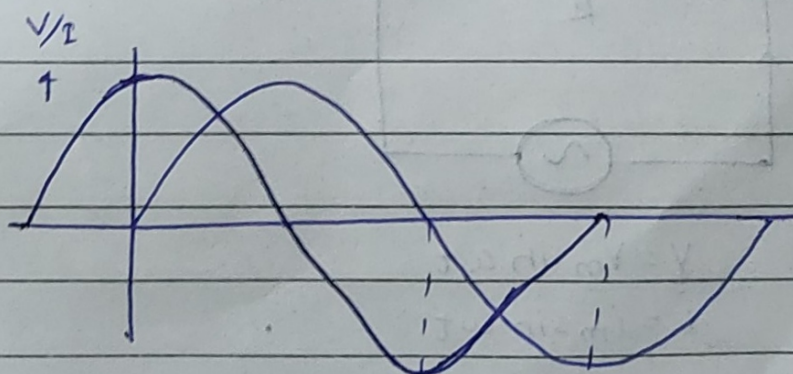
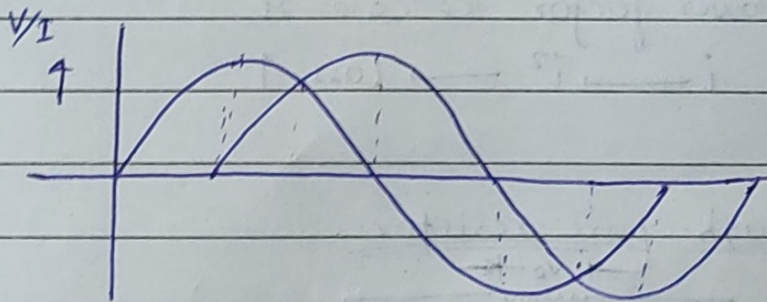
$S = VI \rightarrow \text{VA, kVA}$

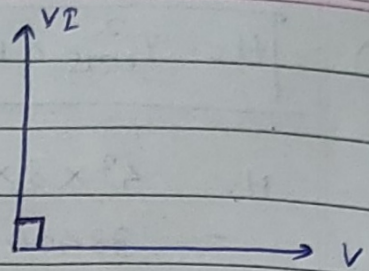
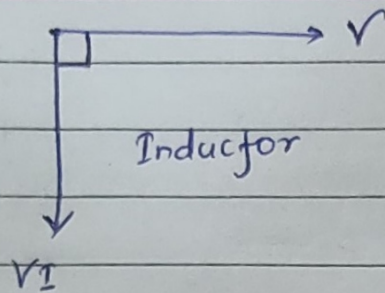
$P_f = \frac{\text{Active Power}}{\text{Apparent Power}} = \cos \phi$

It is cosine of angle b/w voltage and current

$P = VI \cos \phi$

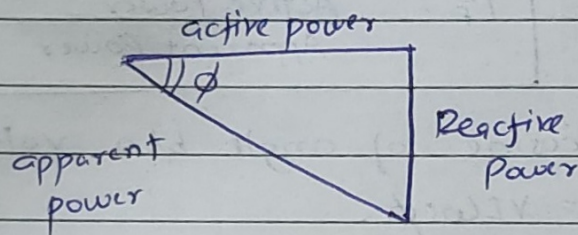
$\cos \phi = \frac{P}{VI}$





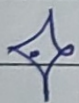
In AC Circuit there are three types of power :-

- ① Active power
- ② Reactive Power
- ③ Apparent Power

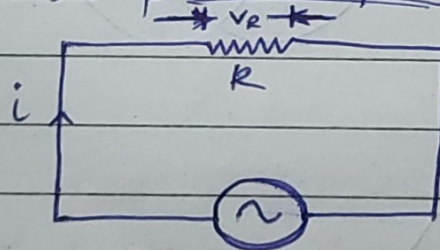


Power factor ke case  $i^2$

$$i \longrightarrow i^2 \longrightarrow \text{Loss} \uparrow$$

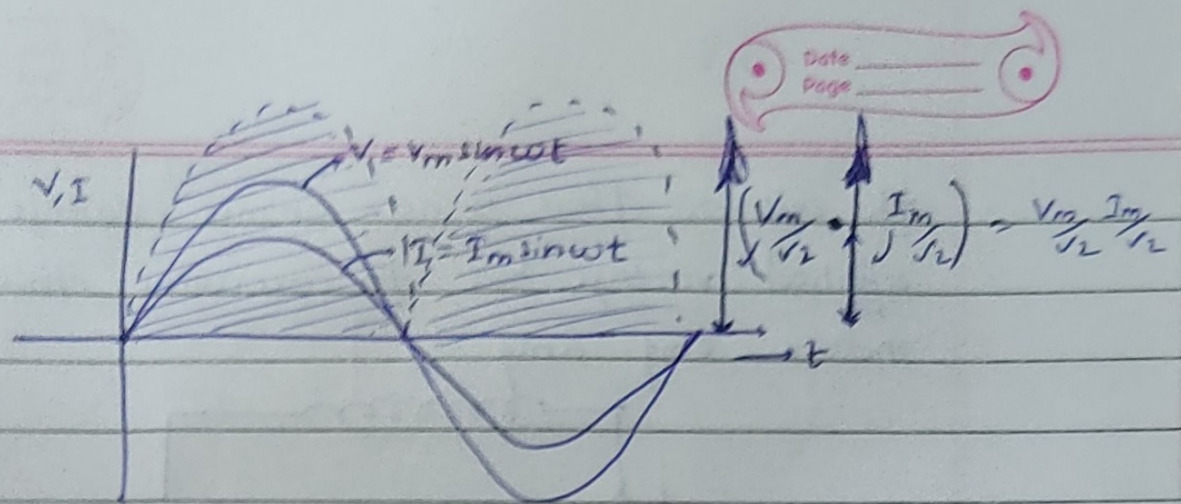


AC through pure resistor



$$V = V_m \sin \omega t$$

$$i = i_m \sin \omega t$$



$$V = IR$$

$$I = I_m \sin \omega t$$

$$V_m \sin \omega t = IR$$

$$\boxed{\frac{V_m}{R} \sin \omega t = I}$$

$$P = VI$$

$$= V_m \sin \omega t \cdot I_m \sin \omega t$$

$$= V_m I_m \sin^2 \omega t$$

$$= V_m I_m \left( \frac{1 - \cos 2\omega t}{2} \right) = \frac{V_m I_m}{2} - \frac{V_m I_m \cos 2\omega t}{2}$$

$\downarrow$   
 0  
 $\downarrow$   
 for one cycle.

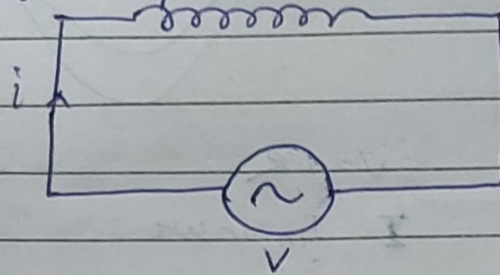
$$= \frac{V_m}{\sqrt{2}} \cdot \frac{I_m}{\sqrt{2}}$$



### AC through pure Inductor.

Whenever an alternating value is applied to a purely inductive coil a back emf is produced due to the self inductance of the coil. The back emf at every step oppose the fall or rise of current through

the coil as there is no ohmic voltage drop the applied voltage has to overcome these self-induce emf only.



$$e_L = -L \frac{di}{dt}$$

$$V + e_L = 0$$

$$V = -e_L$$

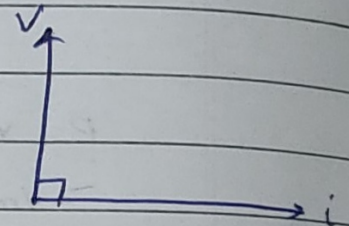
$$= L \frac{di}{dt}$$

$$= L \frac{d(i_m \sin \omega t)}{dt}$$

$$= \omega L I_m \cos \omega t$$

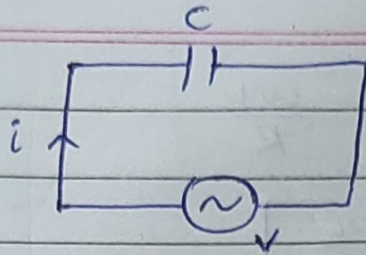
$$= \omega L I_m \sin(\omega t + \frac{\pi}{2})$$

$$V = V_m \sin(\omega t + \frac{\pi}{2})$$



### ❖ AC through pure capacitor

When an alternating voltage is applied to the plates of a capacitor the capacitor is charged first in one direction and then in opposite direction.



$$q = CV$$

$$= CV_m \sin \omega t$$

$$i = \frac{dq}{dt}$$

$$i = \frac{d(CV_m \sin \omega t)}{dt}$$

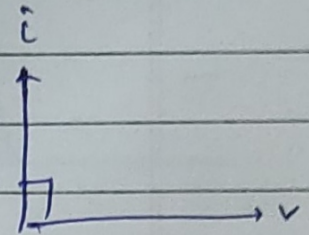
$$= CV_m \cos \omega t$$

$$CV_m \cos \omega t \cdot \omega$$

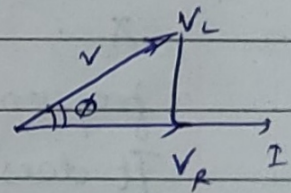
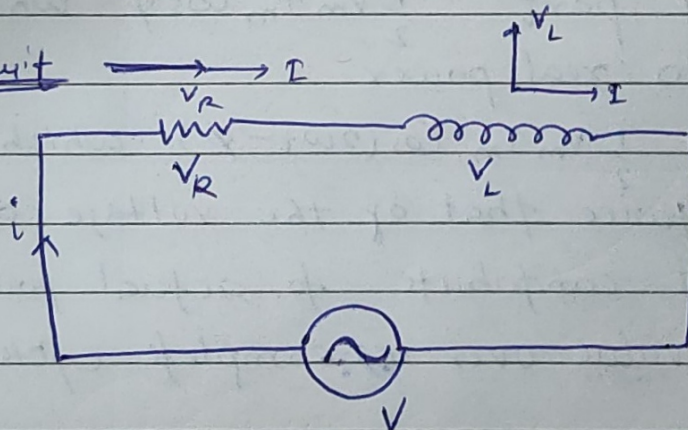
$$\omega CV_m \cos \omega t$$

$$\omega CV_m \sin(\omega t + \pi/2)$$

$$i = i_m \sin(\omega t + \pi/2)$$



RL Circuit



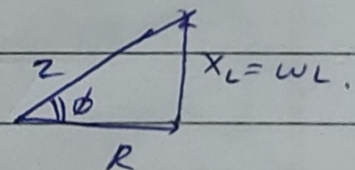
$$V^2 = V_R^2 + V_L^2$$

$$= (IR)^2 + (IX_L)^2$$

$$V = I \sqrt{R^2 + X_L^2}$$

$$\frac{V}{I} = \sqrt{R^2 + X_L^2}$$

$$Z = \sqrt{R^2 + X_L^2}$$



$$\tan \phi = \frac{X_L}{R} = \frac{\omega L}{R}$$

$$\phi = \tan^{-1} \frac{\omega L}{R}$$

$$\begin{aligned} P &= V_m \sin \omega t \cdot I_m \sin (\omega t - \phi) \\ &= \frac{1}{2} V_m I_m [2 \sin \omega t \sin (\omega t - \phi)] \\ &= \frac{1}{2} V_m I_m [\cos \phi - \underbrace{\cos (2\omega t - \phi)}_0] \end{aligned}$$

$$P = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos \phi$$

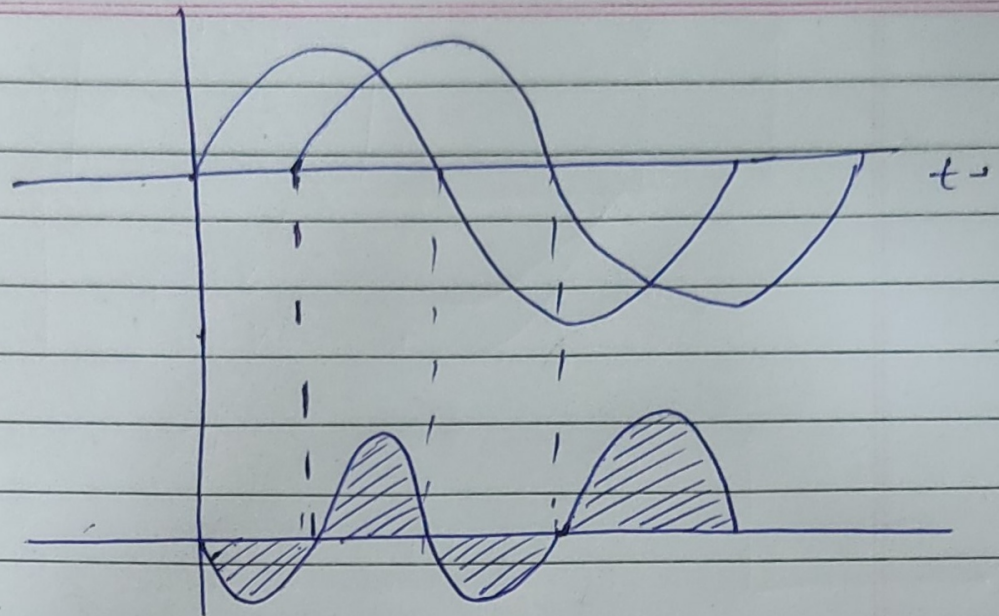
$$P = VI \cos \phi$$

A constant part  $\frac{1}{2} V_m I_m \cos \phi$  which contributes to real power

A  $\frac{1}{2} V_m I_m \cos (2\omega t - \phi)$  which has a frequency twice that of the voltage and current. It does not contribute to actual power, since its average value over a complete cycle is zero

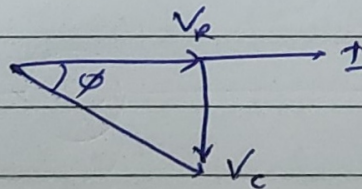
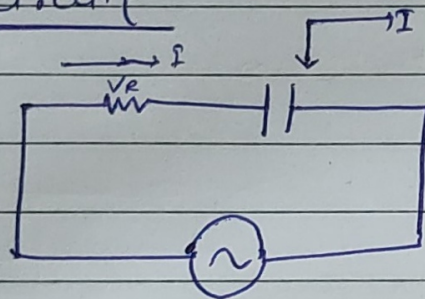
$$\begin{aligned} P &= VI \cos \phi \\ &= VI \left( \frac{R}{Z} \right) \end{aligned}$$

$$\begin{aligned} &= \frac{V}{Z} (IR) \\ &= I^2 R \end{aligned}$$



$R_L \rightarrow$

R-C circuit



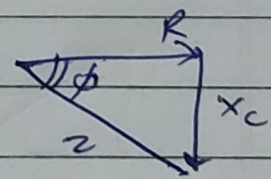
$$V^2 = V_R^2 + V_C^2$$

$$= (IR)^2 + (IX_C)^2$$

$$V = \sqrt{(IR)^2 + (IX_C)^2}$$

$$= I \sqrt{R^2 + X_C^2}$$

$$\frac{V}{I} = \sqrt{R^2 + X_C^2} \Rightarrow Z = \sqrt{R^2 + X_C^2}$$



$$\tan \phi = \frac{X_C}{R}$$

$$\phi = \tan^{-1} \left( \frac{X_C}{R} \right)$$